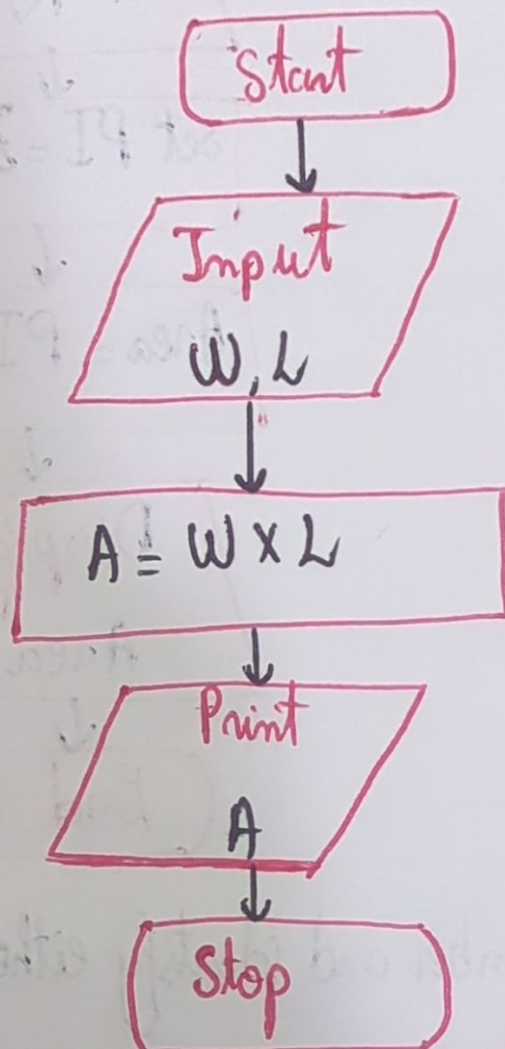
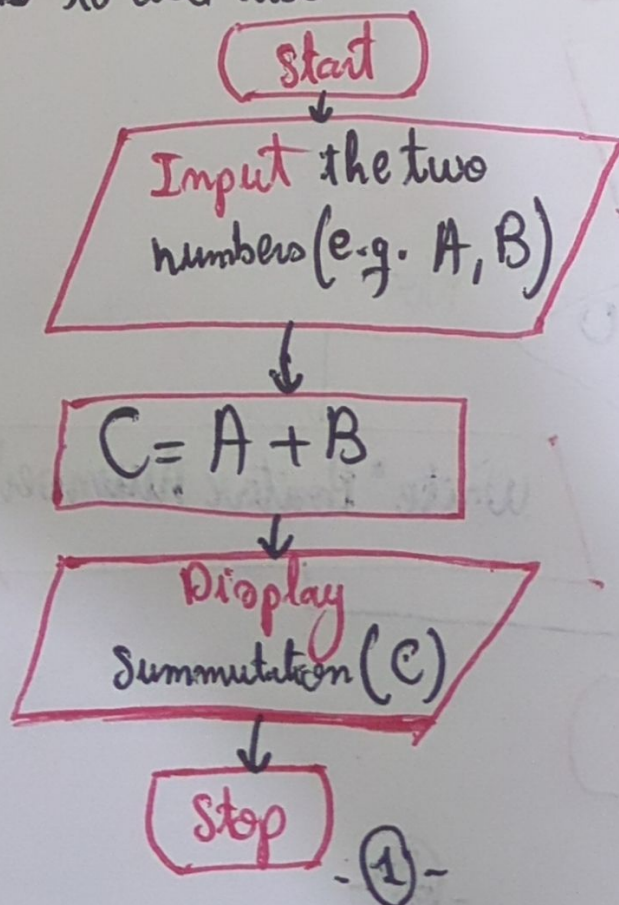


Algorithmic and Data StructuresTutorial Exercises (Series 01) (Solution).Exercise 01 (Flowcharts)

* Draw a flowchart to read the two sides of a rectangle and calculate its area.

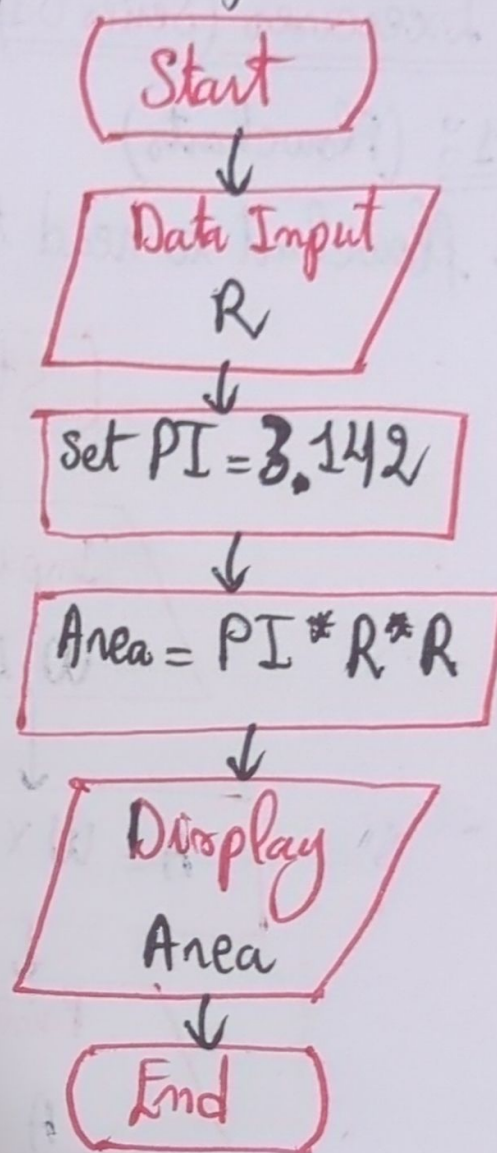
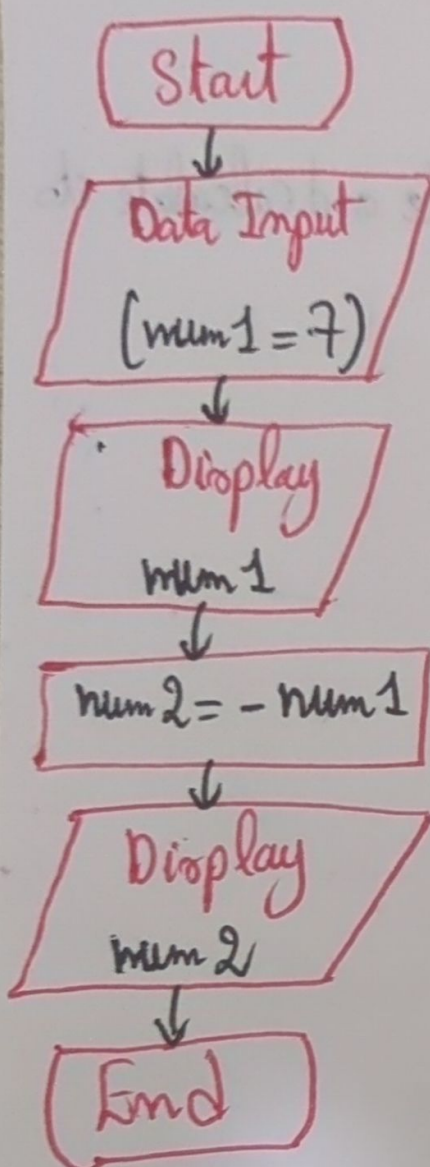


* Draw a flowchart to add two numbers entered by the user.

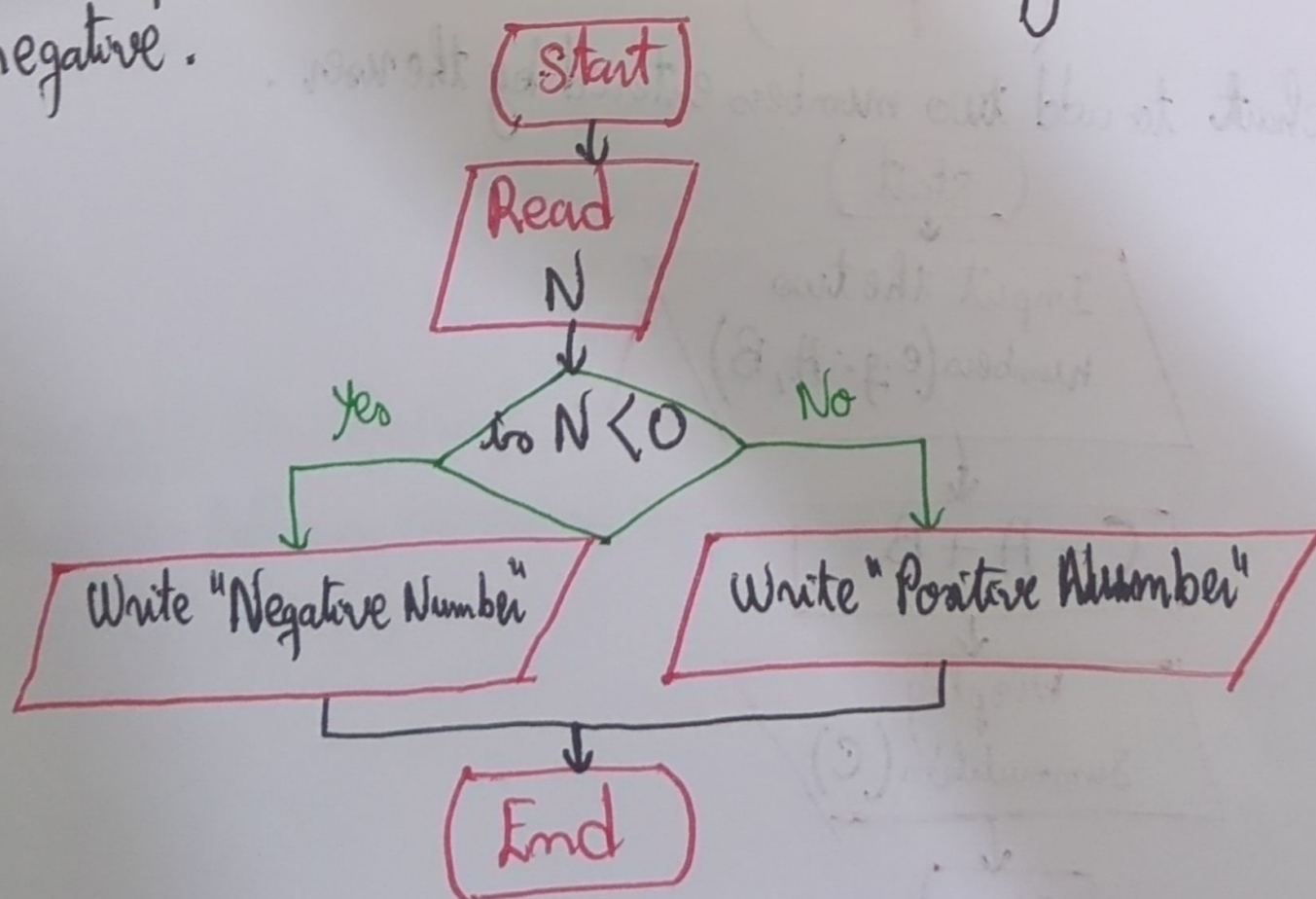


* Draw a flowchart to read a variable $\text{num1} = 7$ and store the negative value of num1 in another variable num2 .

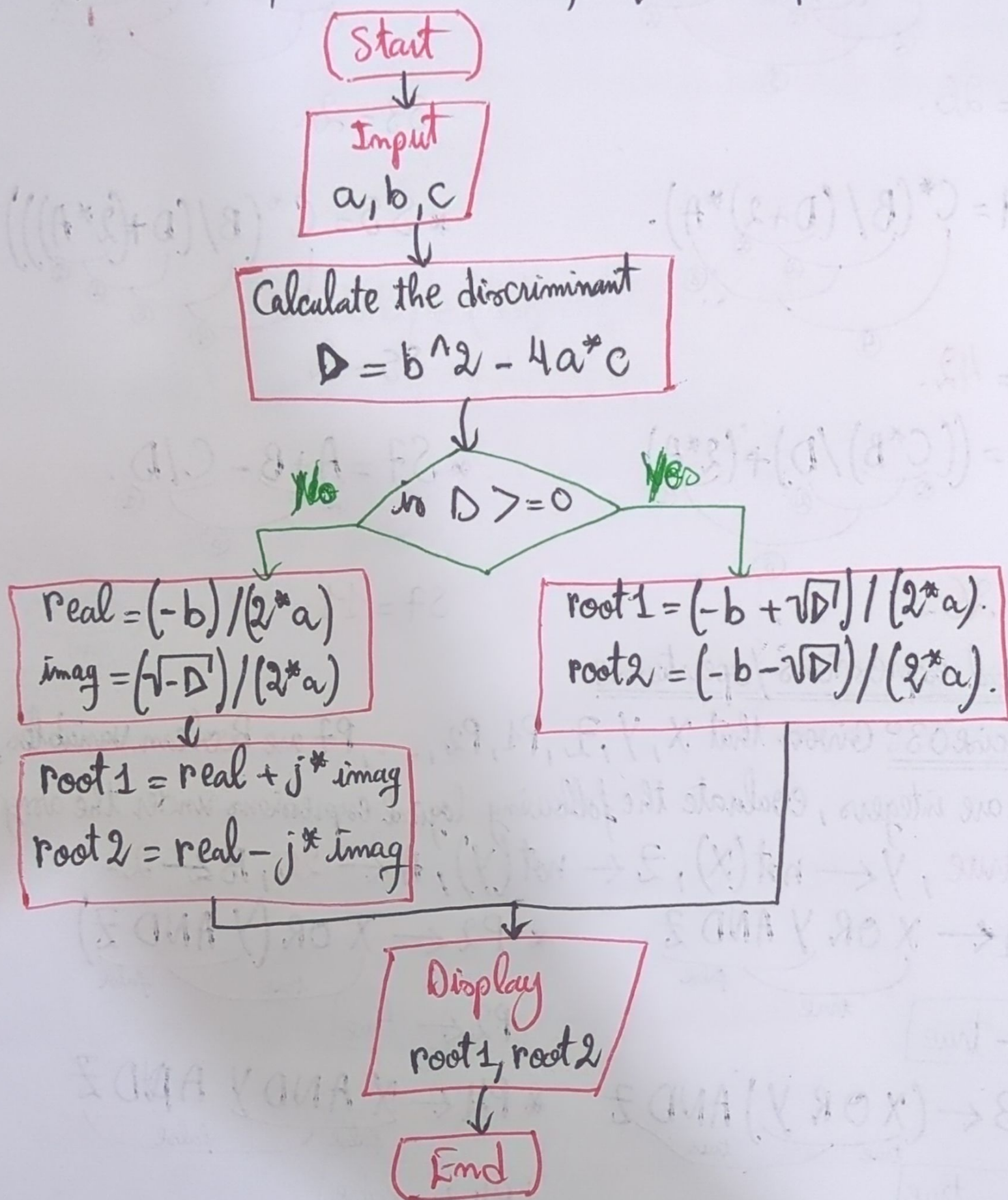
* Draw a flowchart to compute the area of a circle of Radius " R ".



* Draw a flowchart to read a number and identify either the number is positive or negative.



* Draw a flowchart to find all the roots of a quadratic equation $ax^2 + bx + c = 0$



Exercise 02% (Arithmetic)

Evaluate the following numerical expressions given: $A = 6$, $B = 7$, $C = 4$ and $D = 2$

$$* S1 = C * B / D + 2 * A$$

①
②
③
④

$$S1 = 26$$

Operator priority

()	Unary operators: + (unary), - (unary)
NOT	! or not
* / % //	
+ -	
< <= > >=	
== != ; AND ; OR	

$$* S2 = C * (B/D) + 2 * A$$

(2) (1) (2) (3)

$$S2 = 26.$$

$$* S3 = C * B / (D + 2 * A)$$

(1) (2) (3) (1)

$$S3 = 2.$$

$$* S4 = C * (B / (D + 2) * A).$$

(2) (1) (3) (4)

$$S4 = 42.$$

$$* S5 = C * (B / (D + (2 * A)))$$

(4) (3) (2) (1)

$$S5 = 2$$

$$* S6 = ((C * B) / D) + (2 * A)$$

(2) (2) (1) (3)

$$S6 = 26.$$

$$* S7 = A + B - C / D.$$

(2) (3) (1)

$$S7 = 11.$$

Logical expressions / operations

Exercise 03: Given that $X, Y, Z, P1, P2, \dots, P7$ are Boolean variables, and A, B are integers, evaluate the following logical expressions under the assignments $X \leftarrow \text{true}$, $Y \leftarrow \text{not}(X)$, $Z \leftarrow \text{not}(Y)$, $A \leftarrow 20$, $B \leftarrow 2$.

$$* P1 \leftarrow X \text{ OR } Y \text{ AND } Z.$$

true false

$$P1 \leftarrow \text{true}$$

$$* P2 \leftarrow X \text{ OR } (Y \text{ AND } Z)$$

true false

$$P2 \leftarrow \text{true}$$

$$* P3 \leftarrow (X \text{ OR } Y) \text{ AND } Z$$

true true

$$P3 \leftarrow \text{true}$$

$$* P4 \leftarrow X \text{ AND } Y \text{ AND } Z$$

false false

$$P4 \leftarrow \text{false}$$

$$* P5 \leftarrow X \text{ AND } Y \text{ OR } \text{NOT}(Z).$$

false false false

$$P5 \leftarrow \text{false}$$

$$* P6 \leftarrow (A < 4 \text{ OR } P3) \text{ AND } (B + A == 22)$$

false true 22 true

$$P6 \leftarrow \text{true}$$

$$* P7 \leftarrow (A * B + 5) > 45 \text{ AND } \text{NOT } P2.$$

40 45 false

$$P7 \leftarrow \text{false}$$

Exercise 048 Let a, b, c be three Boolean variables. Determine which pairs of the following Boolean expressions are equivalent. For those that are not equivalent, provide an example in which one expression evaluates to true while the other evaluates to false.

De Morgan's laws:

$$\neg(A \wedge B) \equiv (\neg A) \vee (\neg B)$$

$$\neg(A \vee B) \equiv (\neg A) \wedge (\neg B)$$

①. $\neg(a \text{ OR } b)$ with $a \text{ OR } b$ $\leftarrow$ Not Equivalent

$\neg(a) \text{ AND } \neg(b)$ $\leftarrow$ That's the exact opposite of $a \text{ OR } b$.

Example: $a = \text{true}, b = \text{false}$

$\underbrace{\neg(a \text{ OR } b)}_{\text{false}} \quad \text{v.s.} \quad \underbrace{a \text{ OR } b}_{\text{true}}$

②. $\neg(a \text{ AND } b)$ with $(\neg a) \text{ OR } (\neg b)$ $\leftarrow$ Equivalent.

$(\neg a) \text{ OR } (\neg b)$ $\leftarrow$ De Morgan's Law.

③. $\neg(a \text{ OR } (\neg b))$ with $(\neg a) \text{ AND } b$ $\leftarrow$ Equivalent.

$\neg a \text{ AND } b$ $\leftarrow$ De Morgan's Law.

④. $\neg(\neg(\neg a))$ with $(\neg a)$ $\leftarrow$ Equivalent.

$\underbrace{\neg(\neg(\underbrace{\neg a}_a))}_{\neg a}$ $\leftarrow$ Simplify.

⑤. $a \text{ AND } (b \text{ OR } c)$ with $a \text{ AND } b \text{ OR } c$ $\leftarrow$ NOT Equivalent.

$$(a \text{ AND } b) \text{ OR } (a \text{ AND } c) \neq (a \text{ AND } b) \text{ OR } c$$

Examples: $a = \text{false}, b = \text{false}, c = \text{true}$

$\underbrace{(a \text{ AND } b)}_{\text{false}} \text{ OR } \underbrace{(a \text{ AND } c)}_{\text{false}} \quad \text{v.s.} \quad \underbrace{(a \text{ AND } b)}_{\text{false}} \text{ OR } \underbrace{c}_{\text{true}}$

⑥. $a \text{ AND } (b \text{ OR } c)$ with $(c \text{ OR } b) \text{ AND } a \leftarrow$ Equivalent.

Commutative law.

⑦. $a \text{ AND } (b \text{ OR } c)$ with $a \text{ AND } b \text{ OR } a \text{ AND } c \leftarrow$ Equivalent.

Distributive law.

⑧. $a \text{ OR } (b \text{ AND } c)$ with $(a \text{ OR } b) \text{ AND } (a \text{ OR } c) \leftarrow$ Equivalent.

The distributive law

Exercise 058

Consider the following algorithm for solving quadratic equations:

Algorithm Equation2.nd

Var a, b, c, d, sd & real;

$cond0, cond1, cond2, \dots, cond9$ & boolean;

Begin

Write ("Enter a, b, c");

Read (a, b, c);

$d \leftarrow b^2 - 4 * a * c$;

$cond0 \leftarrow (a == 0) \rightarrow$ Equation is not quadratic.

$cond1 \leftarrow (cond0 \text{ AND } b == 0) \rightarrow$ both $a = 0$ and $b = 0$.

$cond2 \leftarrow (cond1 \text{ AND } c == 0) \rightarrow a = 0, b = 0, c = 0 \rightarrow$ Trivial equation.

$cond3 \leftarrow (cond1 \text{ AND } c < > 0) \rightarrow a = 0, b = 0, c \neq 0 \rightarrow$ impossible equation.

$cond4 \leftarrow (cond0 \text{ AND } b < > 0) \rightarrow$ linear equation (ax^2 disappears, but b remains)

$cond5 \leftarrow (a < > 0) \rightarrow$ quadratic case.

$cond6 \leftarrow (cond5 \text{ AND } d < 0) \rightarrow$ quadratic with negative discriminant $\rightarrow$ no real solutions

$cond7 \leftarrow (cond5 \text{ AND } d \geq 0) \rightarrow$ quadratic with real discriminant.

$cond8 \leftarrow (cond7 \text{ AND } d == 0) \rightarrow // \text{ 4 with one double root.}$

$cond9 \leftarrow (cond7 \text{ AND } d > 0) \rightarrow // \text{ 4 with two distinct real roots.}$

* Complete the "if --- then" conditions with the appropriate Boolean variables &

①. $cond2$.

③. $cond4$

⑤. $cond8$.

②. $cond3$.

④. $cond6$

⑥. $cond9$.

⑦.

Instructions, operations & conditions?

Exercise 6

Output $\rightarrow (a, b) = (51, 15)$

This algorithm swaps (exchanges, permute) the values of a and b using the temporary variable c

Permute them circularly example:

If $(a, b, c) = (1, 2, 3)$

After permutation $\rightarrow (a, b, c) = (3, 1, 2)$

Algorithm CircularPermutation

Var

a, b, c, temp : integer

Begin

temp $\leftarrow$ a

a $\leftarrow$ c

c $\leftarrow$ b

b $\leftarrow$ temp

write(a, b, c)

End

What does this algorithm do?

Given three variables a, b and c, give an algorithm to **permute** them circularly, transferring the initial values from "a" to "b", from "b" to "c", and from "c" to "a".

Algorithm Test 1

Var a, b, c : integer;

Begin

a $\leftarrow$ 15;

b $\leftarrow$ 51;

c $\leftarrow$ a; $\leftarrow$ c = 15.

a $\leftarrow$ b; $\leftarrow$ a = 51.

b $\leftarrow$ c; $\leftarrow$ b = 15

write(a, b); $\leftarrow$ a = 51; b = 15

End.

* Write an algorithm that displays the season by entering the month number.

Exercise 7

Algorithm **SeasonFromMonth**

Var

month : integer

Begin

write("Enter the month number (1-12): ")

read(month)

if (month = 12 OR month = 1 OR month = 2) then

write("Winter")

endif

if (month = 3 OR month = 4 OR month = 5) then

write("Spring")

endif

if (month = 6 OR month = 7 OR month = 8) then

write("Summer")

endif

if (month = 9 OR month = 10 OR month = 11) then

write("Autumn")

endif

if (month < 1 OR month > 12) then

write("Invalid month number")

endif

End

Algorithm **EvenOdd**

Var

n : integer

Begin

write("Enter an integer: ")

read(n)

if (n mod 2 = 0) then

write("The number is even")

else

write("The number is odd")

endif

End

Algorithm **MultiplesCheck**

Var

a, b : integer

Begin

write("Enter two integers a and b: ")

read(a, b)

if (b ≠ 0 AND a mod b = 0) then

write("a is a multiple of b")

else if (a ≠ 0 AND b mod a = 0) then

write("b is a multiple of a")

else

write("No multiplicative relationship")

endif

End

* Write an algorithm that reads an integer and then displays whether it is even or odd.

* Write an algorithm that reads two integer variables, "a" and "b", and then indicates whether "a" is a multiple of "b", whether "b" is a multiple of "a", or if there is no multiplicative relationship between "a" and "b".

Exercise 8

Example:

Input: 2024 → divisible by 4 but not by 100 → Leap year

Input: 1900 → divisible by 100 but not by 400 → Not a leap year

Input: 2000 → divisible by 400 → Leap year

Input: 1500 → less than 1584 → Year not valid

Algorithm LeapYearCheck

Var

year : integer

Begin

write("Enter a year: ")

read(year)

if (year > 1584) then

if ((year mod 4 = 0 AND year mod 100 ≠ 0) OR (year mod 400 = 0)) then

write("Leap year (366 days)")

else

write("Not a leap year (365 days)")

endif

else

write("Year not valid (must be > 1584)")

endif

End

* A year is said to be a leap year (it has 366 days) if it greater than 1584,
and
=> The year is divisible by 4 and not divisible by 100.
=> OR the year is divisible by 400.
Otherwise, the year is not a leap year (it has 365 days).
Write an algorithm that reads a year > 1584 and then indicates whether
it is a leap year.

* Write an algorithm that calculates the square of the maximum of two integers provided by the user.

Exercise 9

Algorithm SquareMax Var a, b, max, square : integer Begin write("Enter two integers a and b: ") read(a, b) if (a > b) then max ← a else max ← b endif square ← max * max write("The square of the maximum is: ", square) End	Algorithm Sort2_IfOnly Var a, b, temp : integer Begin write("Enter two integers a and b: ") read(a, b) if (a > b) then temp ← a a ← b b ← temp endif write("Order: ", a, " < ", b) End
Algorithm Sort2_IfAnd Variables a, b : integer Begin write("Enter two integers a and b: ") read(a, b) if (a ≤ b AND true) then write("Order: ", a, " < ", b) endif if (b < a AND true) then write("Order: ", b, " < ", a) endif End	Algorithm Sort2_IfElseAnd Variables a, b : integer Begin write("Enter two integers a and b: ") read(a, b) if (a ≤ b AND true) then write("Order: ", a, " < ", b) else write("Order: ", b, " < ", a) endif End

* Write an algorithm that sorts two integers "a" and "b" provided by the user in ascending order & * Propose three solutions:

- ①. Using only **if** --- **then** conditions.
- ②. Using **if** --- **then** conditions and the **AND** operator.
- ③. Using **if** --- **else** conditions and the **AND** operator.

$$b > a$$

* For homework: do the same exercise with three different integers.

Exercise 10

Algorithm PhotocopyCost

Var

n, cost : integer

Begin

write("Enter the number of photocopies: ")

read(n)

if (n <= 10) then

cost ← n * 3

else if (n <= 30) then

cost ← (10 * 3) + (n - 10) * 2

else

cost ← (10 * 3) + (20 * 2) + (n - 30) * 1

endif

write("Total cost = ", cost, " dinars")

End

made 15 copies.
Let's say the user ~~are~~ already fixed at
* The first "10" are already fixed at $(10 * 3)$
* The extra ones are only $(15 - 10) = 5$ copies
 $5 * 2 \leftarrow (N - 10) * 2$

* A reprography shop charges "3" dinars for the first ten photocopies, 2 dinars for the next twenty, and 1 dinar for each additional photocopy. Write an algorithm that asks the user for the number of photocopies made and then displays the total cost.

→ Next 20 copies (i.e from the 11th to the 30th) → 2 dinars.

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